

# Microplastic Polymer Analysis Report

|              |                                      |
|--------------|--------------------------------------|
| Report ID    | f2bc7430-7192-40dd-bd28-b09cd7e77bf4 |
| Generated On | 18 May 2026 16:09                    |
| Status       | Completed                            |
| Sample ID    | HDPEFTIR001                          |

## Prediction Results

|                  |                 |
|------------------|-----------------|
| Detected Polymer | HDPE            |
| Confidence Score | 100.00%         |
| Model Used       | Peak Identifier |

## FTIR Spectral Analysis

- 2915  $\text{cm}^{-1}$  → CH asymmetric stretching
- 1470  $\text{cm}^{-1}$  → CH bending

## Confidence Interpretation

■■■■■■■■■■ 100.00%

## Scientific References

- [1] Harper, C. Handbook of Plastics Technologies.
- [2] Thompson, R. Plastic debris in oceans.
- [3] Barnes, D. Accumulation of plastics.
- [4] Andrady, A. Environmental impacts of plastics.
- [5] Jung, M. FTIR analysis of HDPE.

## AI Generated Summary

The analyzed sample was identified as HDPE with a confidence score of 100.00%. The FTIR spectral characteristics strongly support the presence of HDPE polymer chains. The detected functional groups and

absorption bands are consistent with known spectral signatures of HDPE. The analysis indicates a reliable classification suitable for environmental and material identification studies.